

CS & IT ENGINEERING



Operating System

Memory Management

Lecture -3

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Recap of Previous Lecture



Doubt?

Topic

**Non-Contiguous Memory
Management Technique**

Topic

Paging

Topic

Address Translation



Topics to be Covered



Topic

TLB

Topic

TLB Mapping

Topic

Segmentation

#Q. Consider a paged memory system where the process size is 16MB and main memory size is 4GB. The page size is 2KB.

1. Number of pages in process? 2^{13}
2. Number of frames in main memory? 2^{21}
3. Number of bits for page number? 13
4. Number of bits for frame number? 21
5. Number of entries in page table? 2^{13}
6. Page table size? $2^{13} * 21$ bits

#Q. Consider a paged memory system where the process size is 128MB and main memory size is 2GB. The page size is 1KB.

1. Number of pages in process? 2^{17}
2. Number of frames in main memory? 2^{21}
3. Number of bits for page number? 17
4. Number of bits for frame number? 21 bits
5. Number of entries in page table? 2^{17}
6. Page table size? $2^{17} * 21 \text{ bits}$

#Q. Consider a paged memory system where the logical address is 25 bits and physical address is 33 bits. The page size is 4KB.

1. Number of pages in process? 2^{13}
2. Number of frames in main memory? 2^{21}
3. Number of bits for page number? 13
4. Number of bits for frame number? 21
5. Number of entries in page table? 2^{13}
6. Page table size? $2^{13} * 21 \text{ bits}$

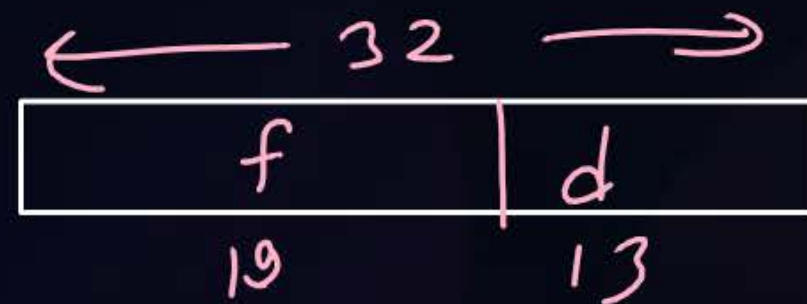
[MCQ]

[GATE-2015]



#Q. A computer system implements 8 kilobyte pages and a 32-bit physical address space. Each page table entry contains a valid bit, a dirty bit, three permission bits, and the translation. If the maximum size of the page table of a process is 24 megabytes, the length of the logical address supported by the system is 36 bits?

LA



Page table size = no of pages * 1 P.T.E size

$$24 \text{ M bytes} = x * 24 \text{ bits}$$

~~$$24 \text{ M} * 8 \text{ bits} = x * 24 \text{ bits}$$~~

$$x = 8 \text{ M} = 2^{23} \Rightarrow p = 23 \text{ bits}$$

$$\begin{aligned} \text{P.T.E. size} &= 19 + 5 \text{ bits} \\ &= 24 \text{ bits} \end{aligned}$$

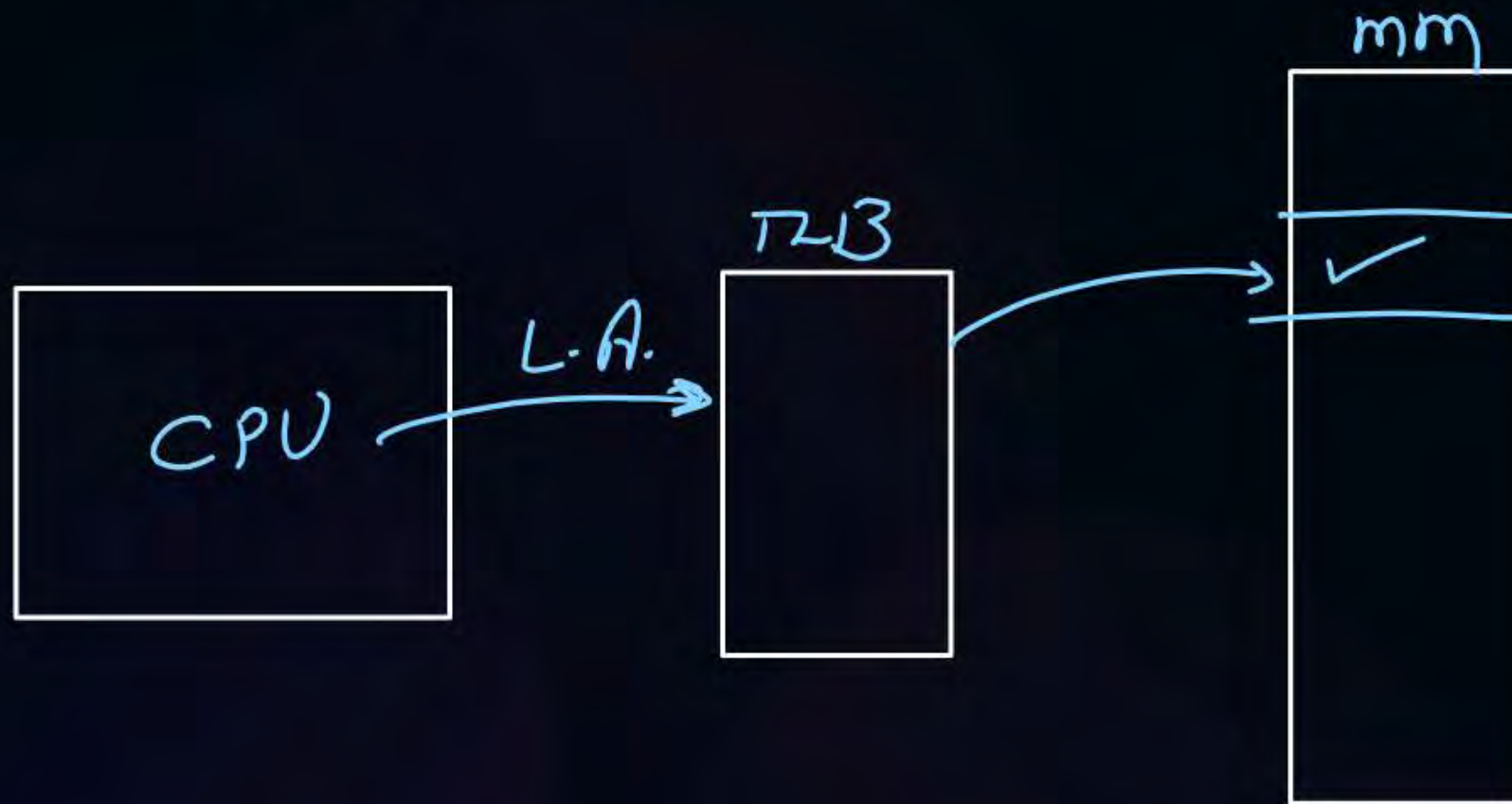


Topic : Paging

TLB (Translation Lookaside Buffer)

used to store some of the frequently and recently accessed p.t. entries.

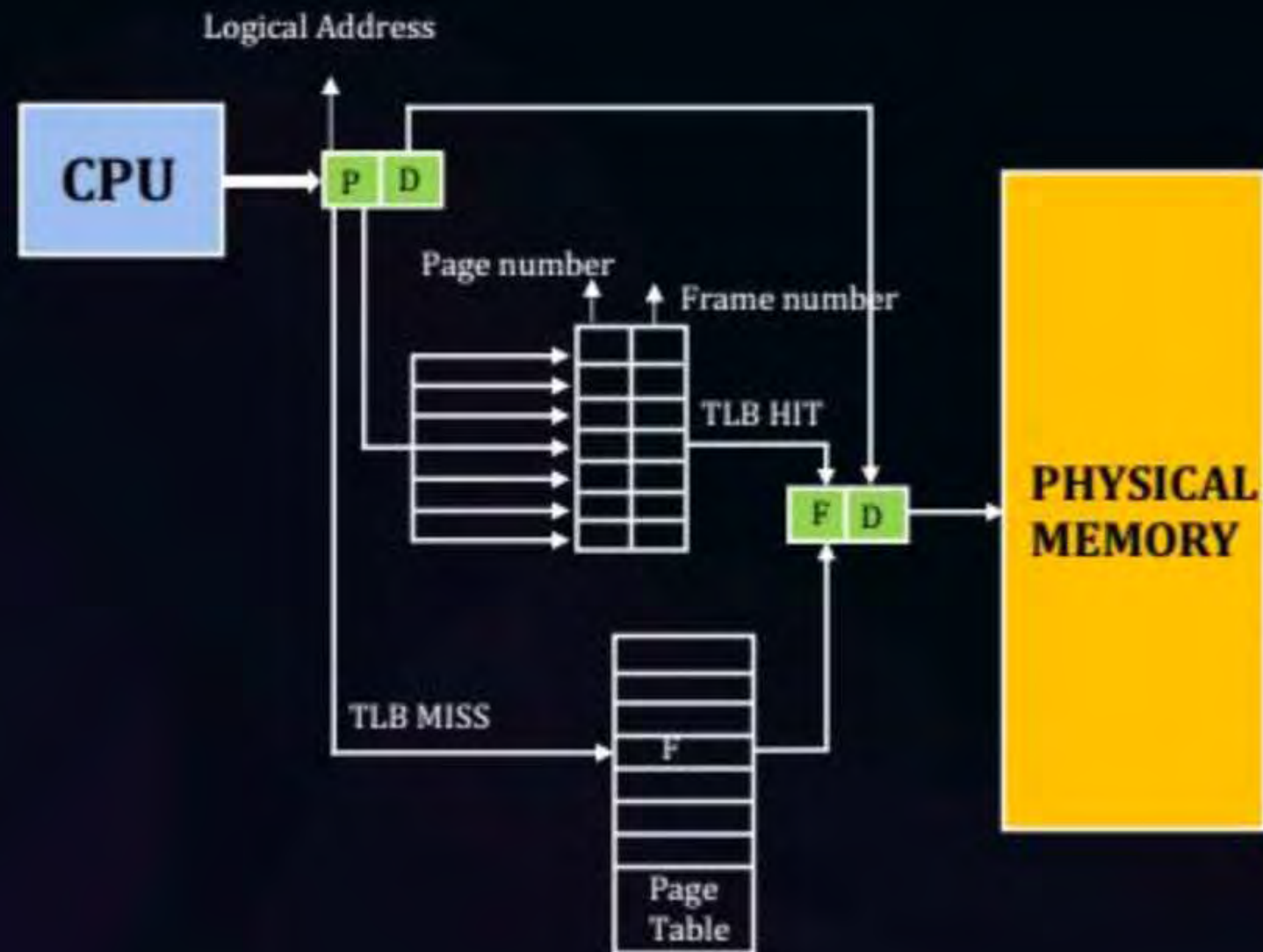
hardware





Topic : Paging

TLB (Translation Lookaside Buffer)



$$\begin{aligned}
 \text{E.M.A.T. (with TLB)} &= H * \left(\underbrace{t_{TLB}}_{\text{translation}} + \underbrace{t_{mm}}_{\text{content}} \right) + (1-H) \left(\underbrace{t_{TLB}}_{\substack{\text{to} \\ \text{check} \\ \text{miss}}} + \underbrace{2t_{mm}}_{\substack{\text{one for P.T.} \\ \text{one for content}}} \right) \\
 &\quad \text{or}
 \end{aligned}$$

$$= t_{TLB} + t_{mm} + (1-H)t_{mm}$$

ques) $t_{TLB} = 10 \text{ ns}$

$$t_{mm} = 200 \text{ ns}$$

$$H = 90\%$$

$$\begin{aligned} \text{EMAT with TLB} &= 10 + 200 + 0.1 * 200 \\ &= 230 \text{ ns} \end{aligned}$$

$$\begin{aligned} \text{EMAT without TLB} &= 2 * 200 \\ &= 400 \text{ ns} \end{aligned}$$

speed up or performance
gain of TLB as
compared not using

$$\text{TLB} = \frac{400 \text{ ns}}{230 \text{ ns}}$$

$$= 1.74$$



Topic : Paging



How TLB Stores Entries?



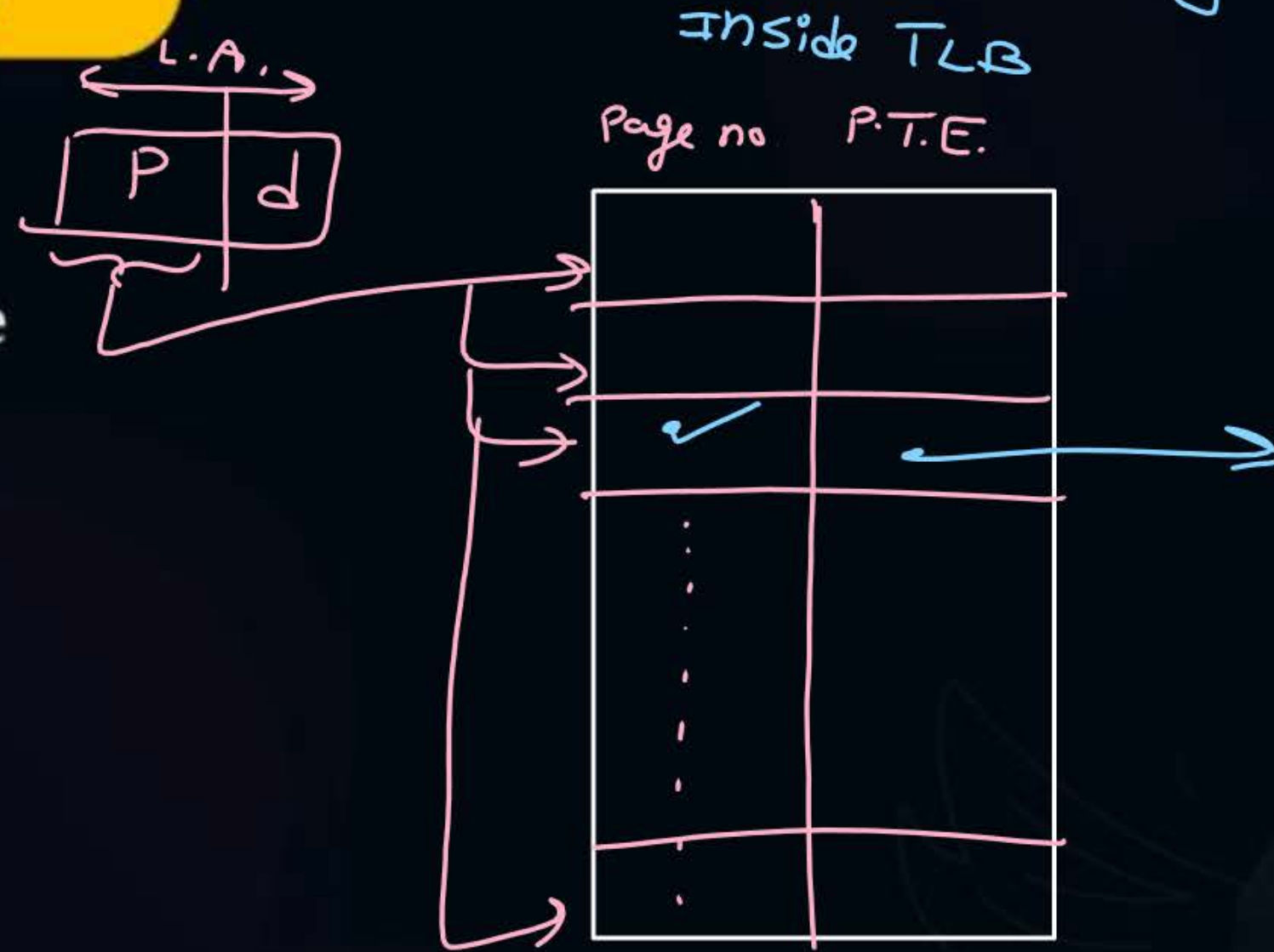
Topic : Paging



fully associative

TLB Mapping

1. Fully Associative
2. Direct
3. Set-Associative

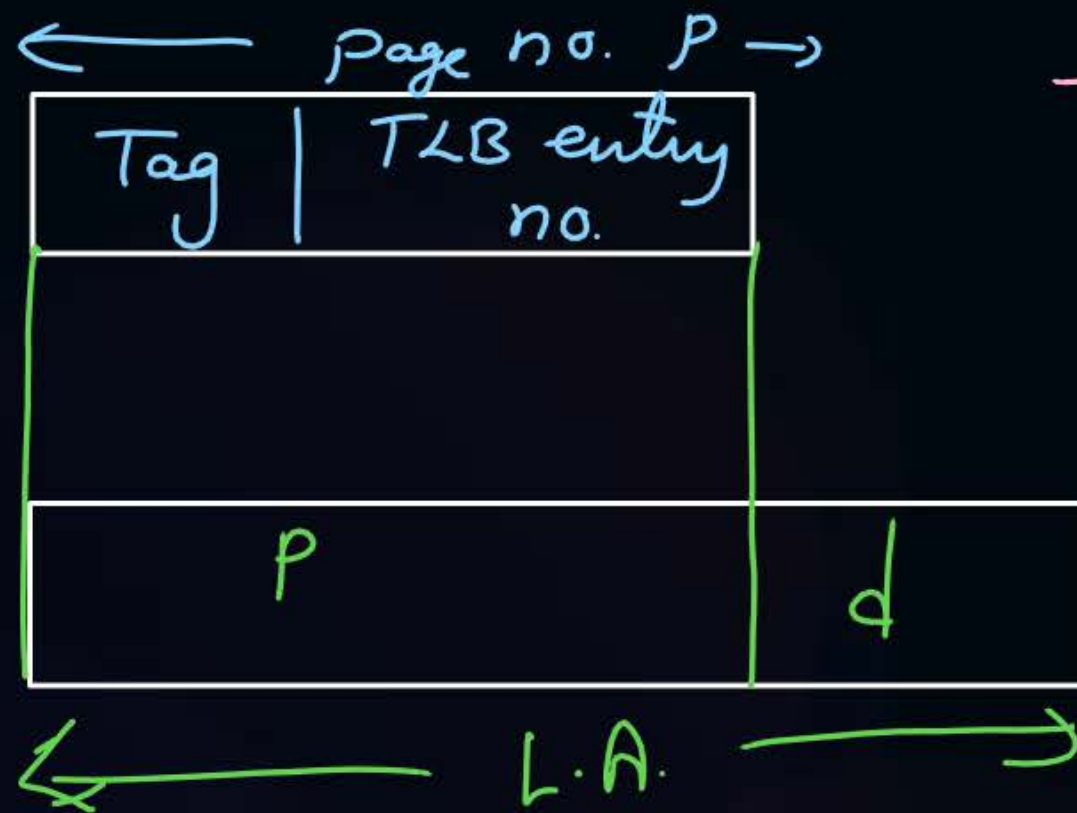




Topic : Paging



TLB Mapping: Direct

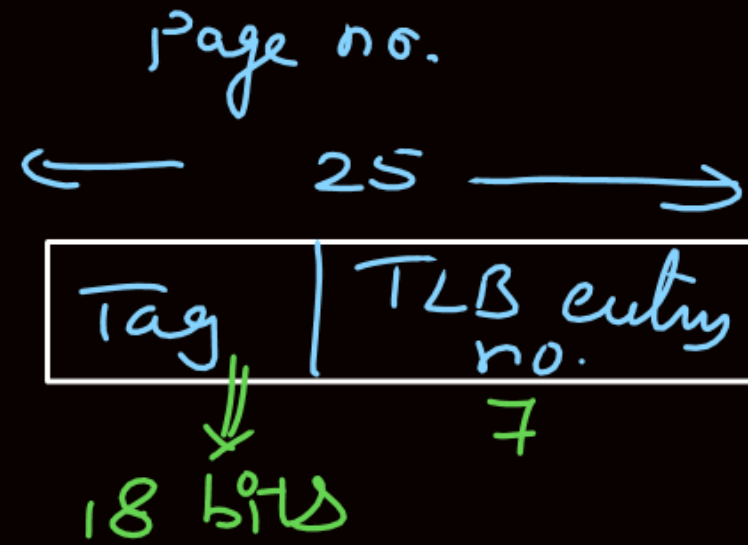


$$\text{no of entries in TLB} = \frac{\text{TLB Size}}{\text{P.T.E. Size}}$$

if TLB can store entries

16	4 bits
32	5 bits
2^x	x bits

ques)



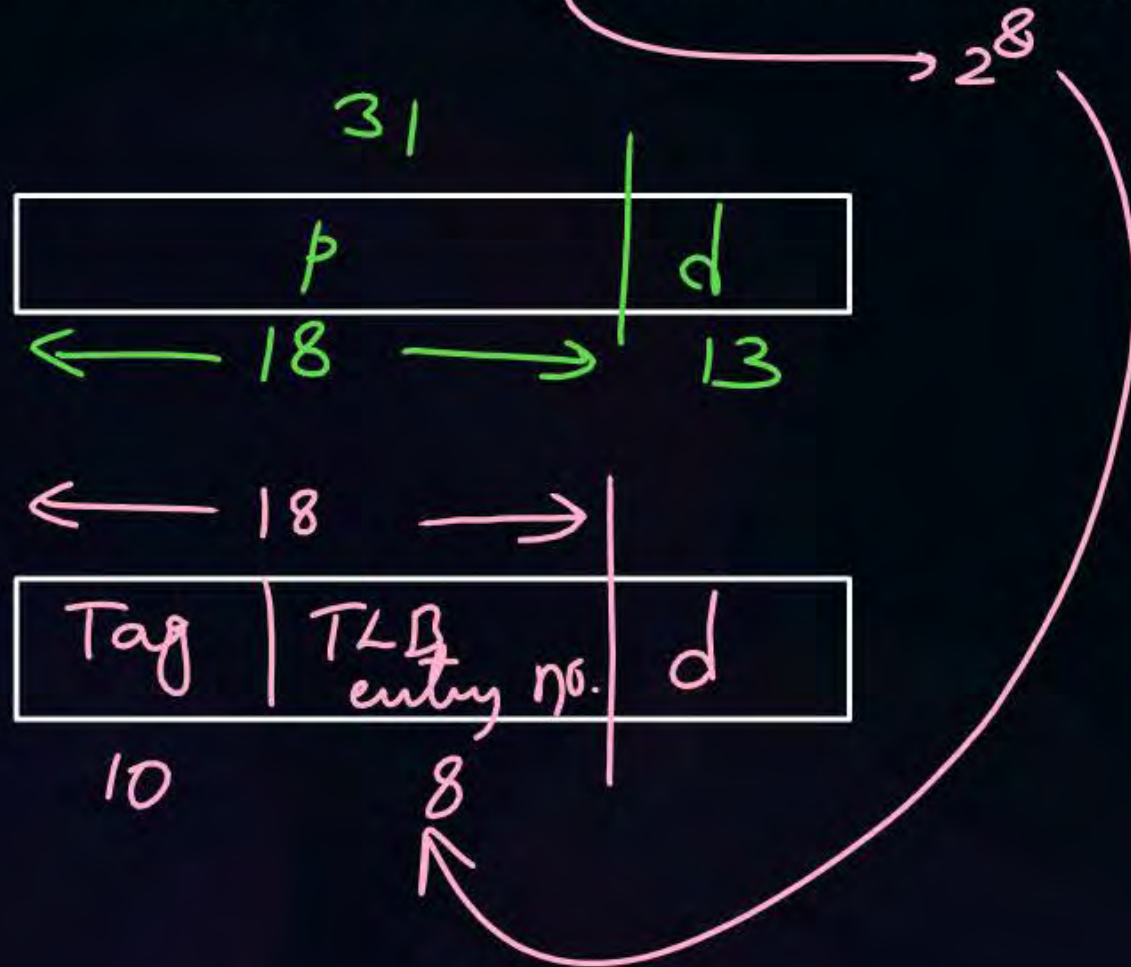
TLB Can store = 128 entries

TLB entry no. = 7 bits



Topic : Question

#Q. A computer system implements a 31-bit virtual address, page size of 8 kilobytes, and a 256-entry translation look-aside buffer (TLB) organized as direct mapped. The minimum length of the TLB tag in bits is 10?



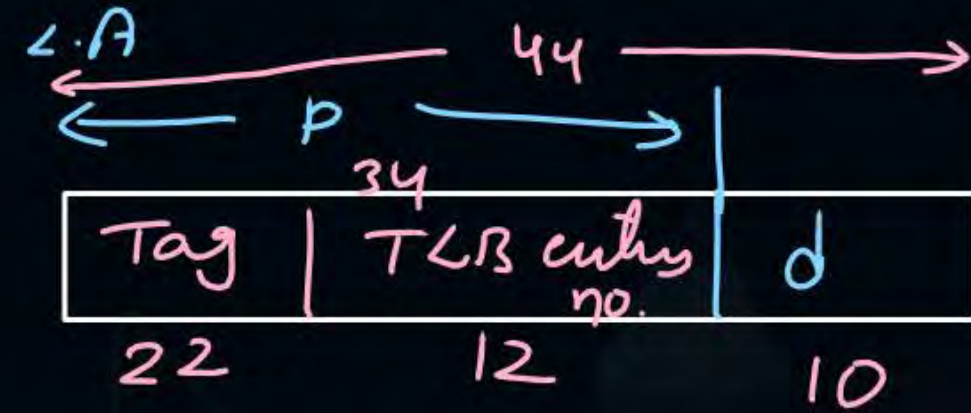


Topic : Question

$2^{10} B$

#Q. A computer system implements a 44-bit virtual address, page size of 1 kilobytes, and a 16KB look-aside buffer (TLB) organized as direct mapped. Each page table entry is of 4bytes. The minimum length of the TLB tag in bits is 22?

$$\begin{aligned}\text{no. of entries in TLB} &= \frac{16KB}{4B} \\ &= 4K \\ &= 2^{12}\end{aligned}$$

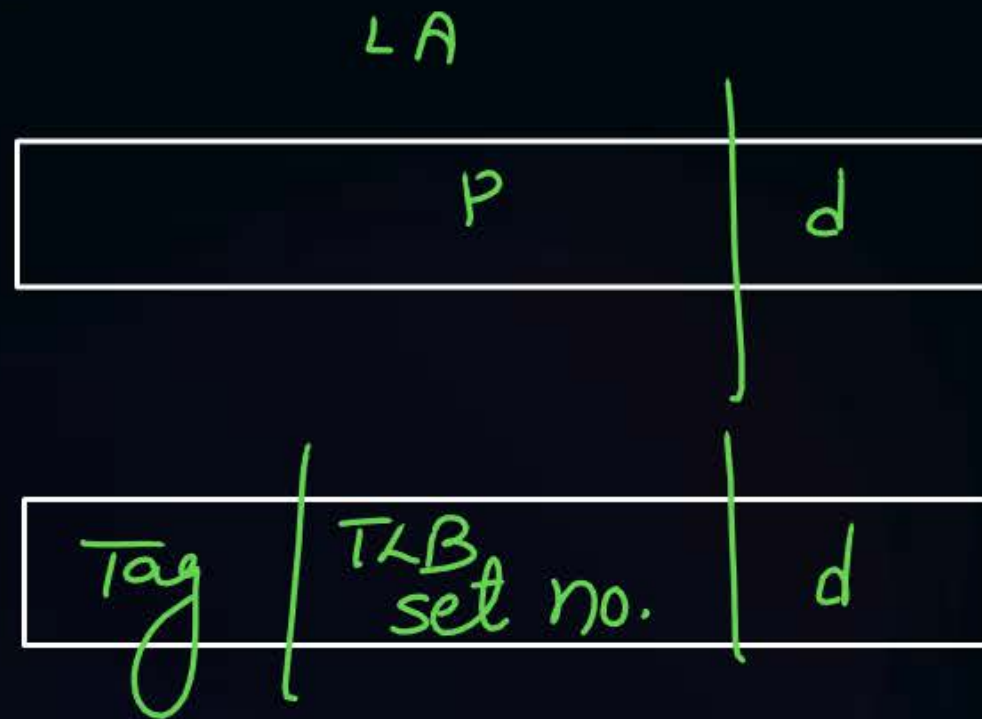




Topic : Paging



TLB Mapping: Set Associative



$$\text{no. of sets in TLB} = \frac{\text{no. of entries in TLB}}{\text{associativity}}$$

no of bits in TLB set no.

$$= \log_2 (\text{no. of sets in TLB})$$

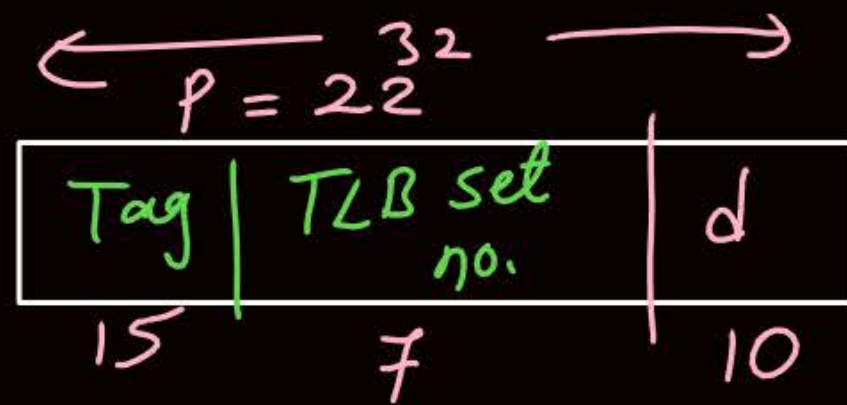
ques 2-way set ass. TLB

TLB can store 256 entries

LA = 32 bits

Page size = 1KB = 2^{10} B

Tag = 15 bits?



$$\text{no. of sets in TLB} = \frac{256}{2}$$

$$= 128$$

$$= 2^7$$

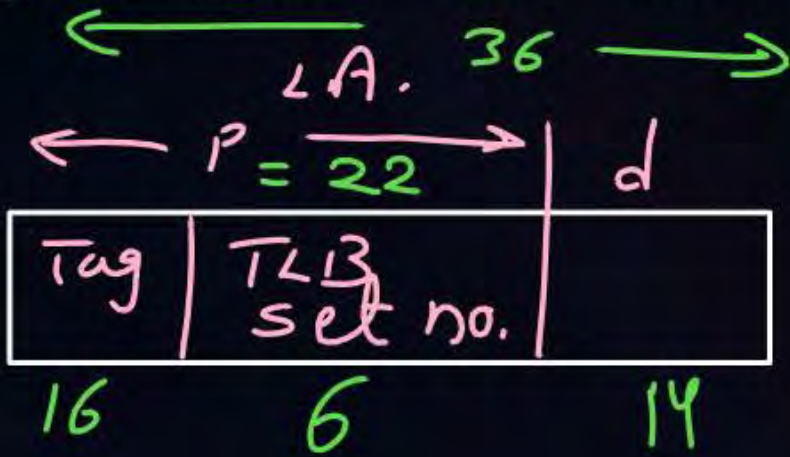
↓

set no. = 7 bits



Topic : Question

#Q. A Computer system implements a 36-bit virtual address, page size of 16 Kbytes and a 256 – entry translation look-aside buffer (TLB) organized into 64 sets each having four ways. Assume that the TLB tag does not store any process id. The minimum length of the TLB tag in bits is 16.

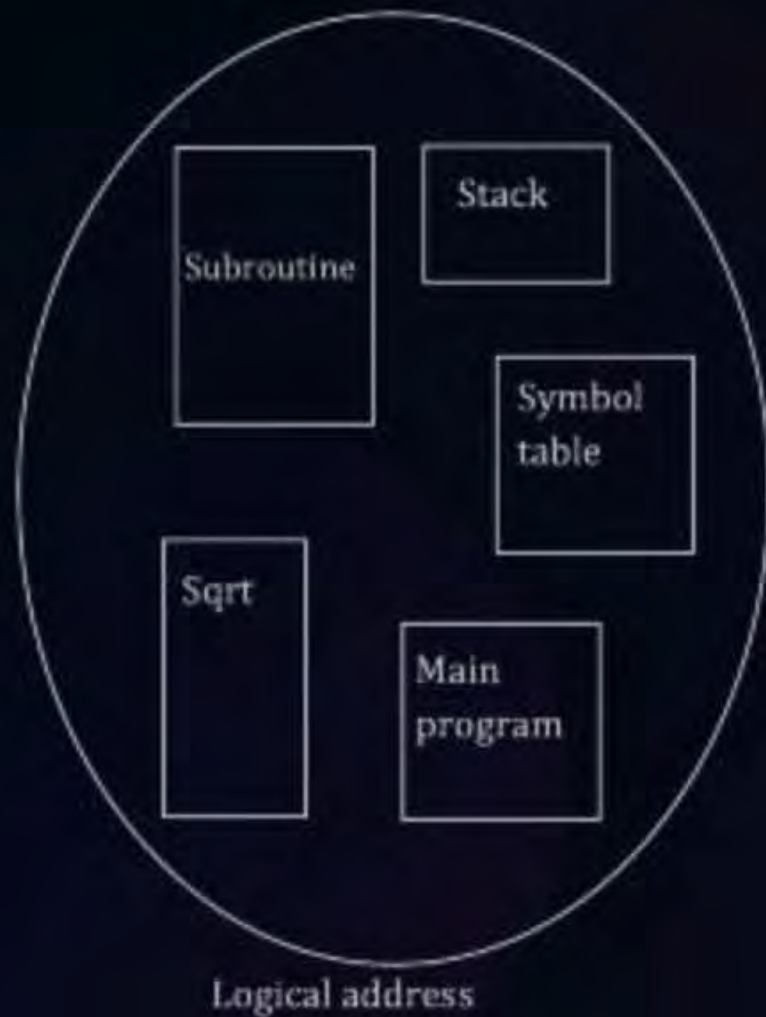


4-way
64 sets = 2^6

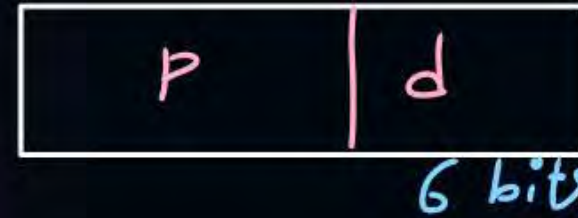


Topic : Segmentation

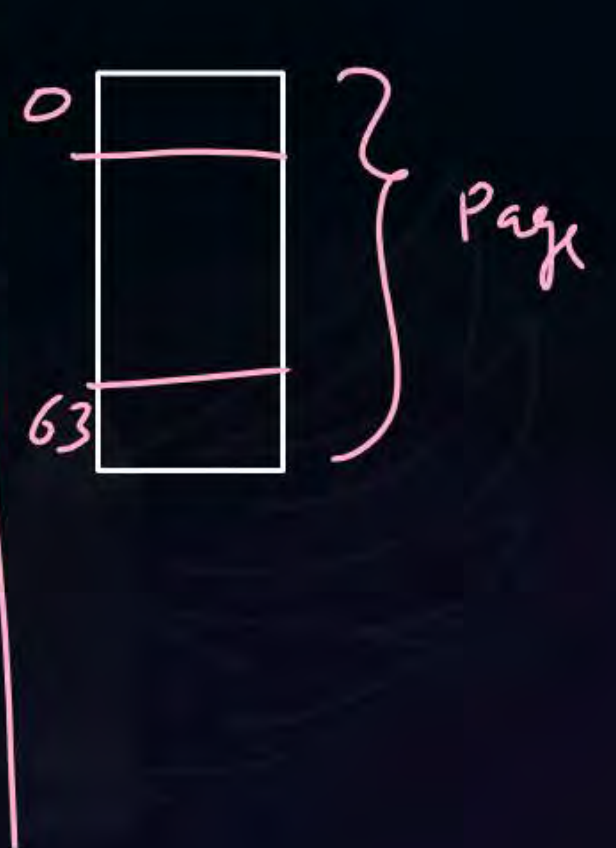
- Divide Process in logically related partitions (Segments) of variable size
- Segments are scattered in physical memory



ex:- Paging
1A

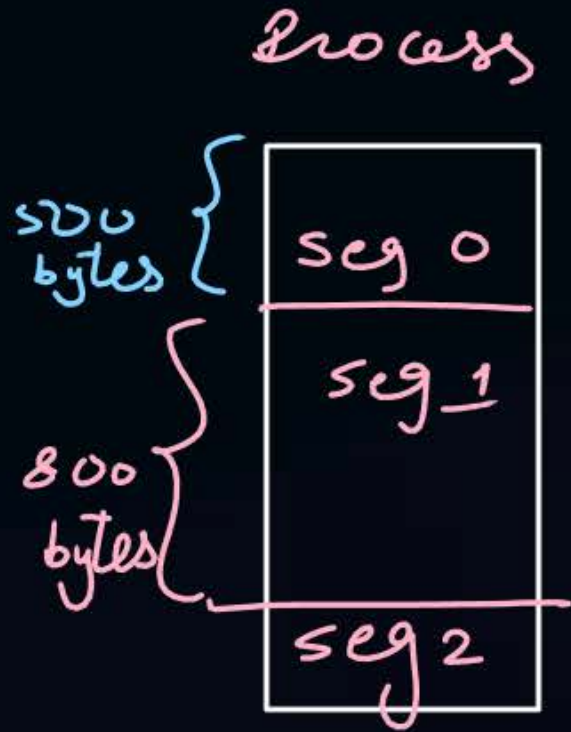


Page size = 64 bytes $\Rightarrow$ in each page
byte 0 to byte 63





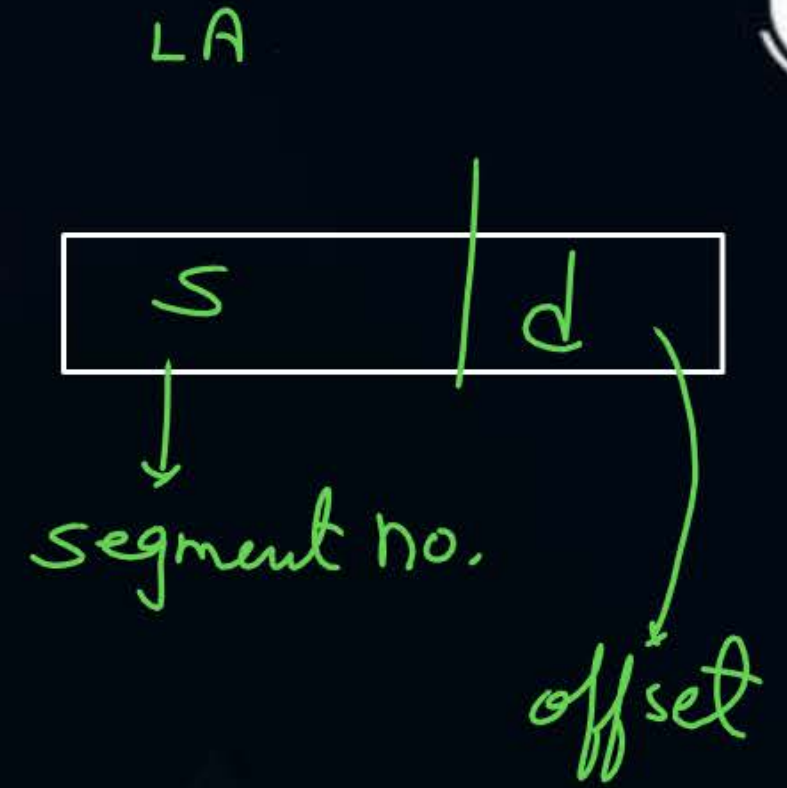
Topic : Segmentation



segment table

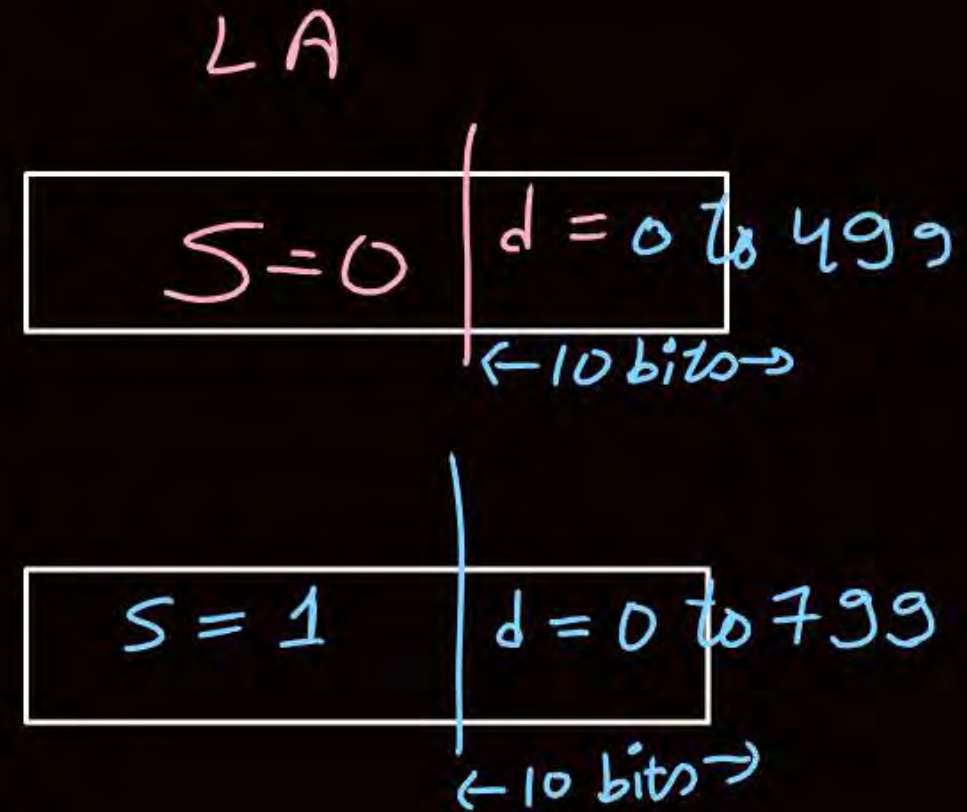
0	1500
1	500
2	2200

starting
add. of segment



no. of bits in offset (d)
 $= \log_2 (\text{largest seg. size possible})$

if any segment size = 500 bytes $\Rightarrow$ possible offset = 0 to 499
(Seg 0)





Topic : Segmentation

example:-

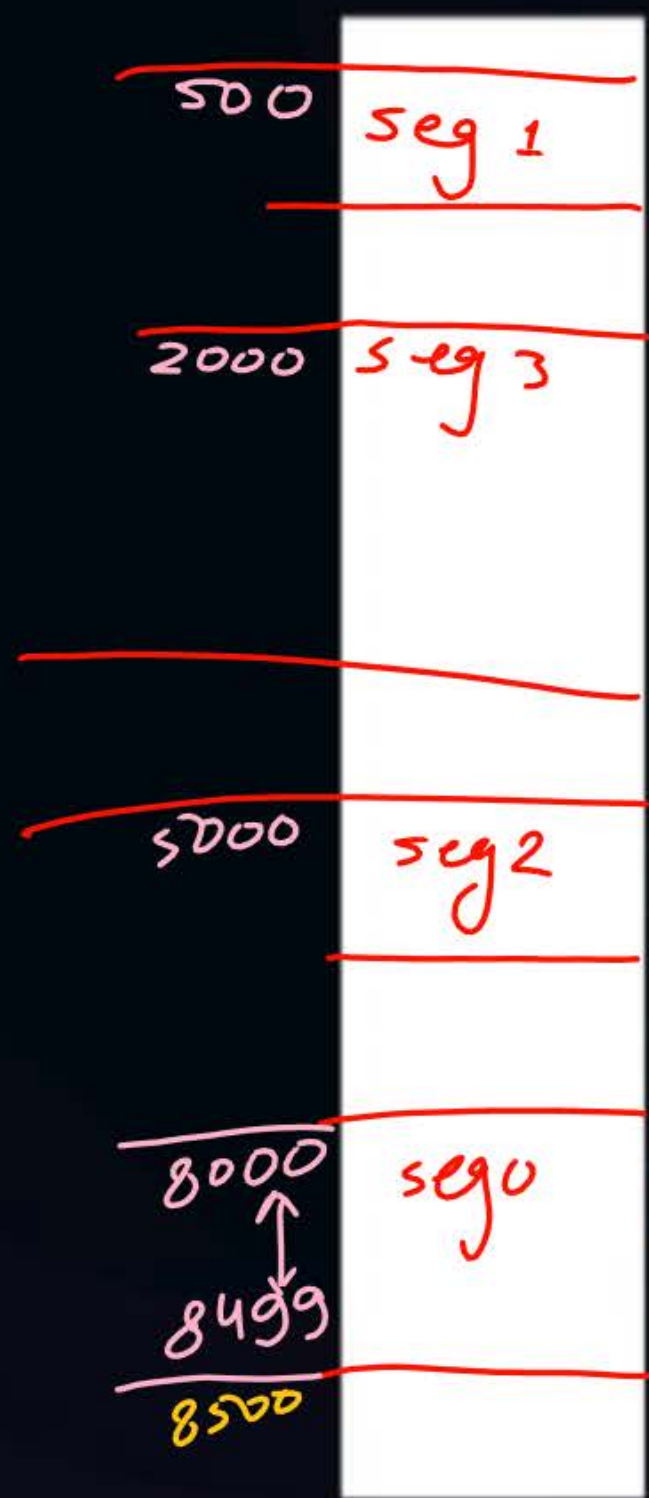
Process
seg 0
seg 1
seg 2
seg 3

	Segment Table	
00	8000	500
01	500	200
10	5000	300
11	2000	800

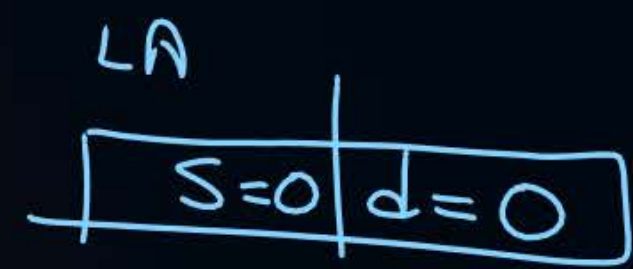
base
(starting
add. of
seg.)

limit
(size of
each segment)

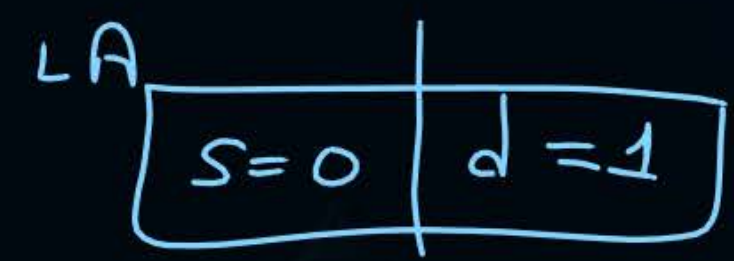
Physical Memory



byte 0
byte 1



P.A. = 8000



PA = 8001

$$P.A = base + d$$

LA $\Rightarrow$ S=0, d=500

P.A = 8500

$\downarrow$ seg. fault



Topic : Segmentation



Segment Table

00	8000	500
01	500	200
10	5000	300
11	2000	800

base

limit

Segment number	Offset	Physical Address
1	143	$500 + 143 = 643$
2	334	fault
3	739	$2000 + 739 = 2739$
0	444	$8000 + 444 = 8444$



Topic : Segmentation

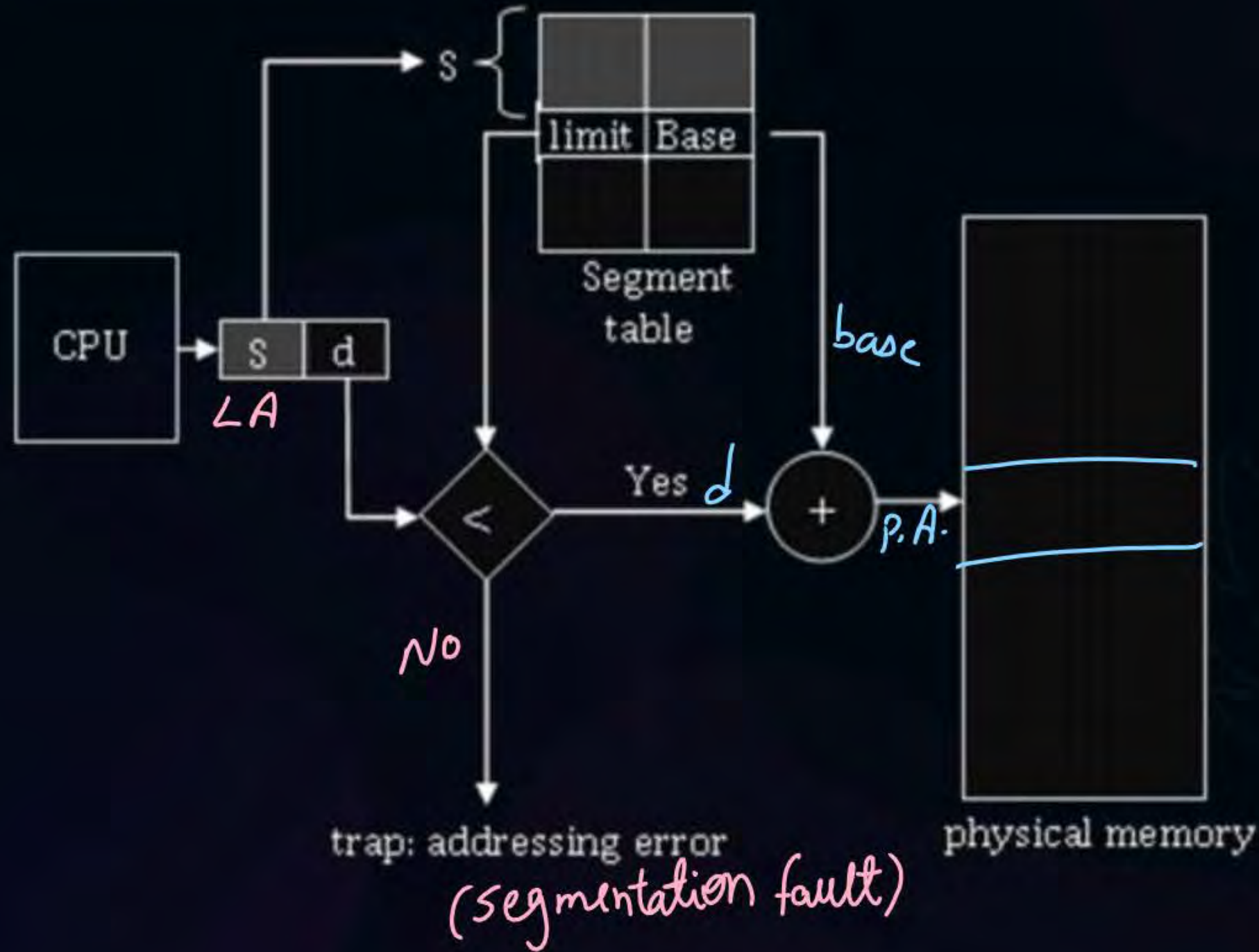


- Size of segment can vary, so along with base, keep limit information also
- Limit defines ~~the~~ number of ~~words~~ *bytes* within the segment





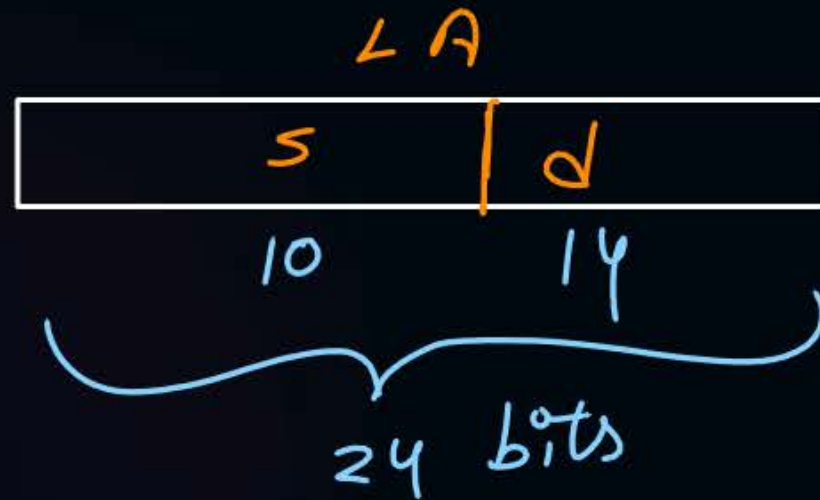
Topic : Segmentation



[NAT]



- #Q. Maximum segment size = 16KB = 2^{14} B $\Rightarrow d = 14$ bits
Number of segments in process = 2^{10} $\Rightarrow s = 10$ bits
Logical address = 24 bits ??





Topic : Segmentation



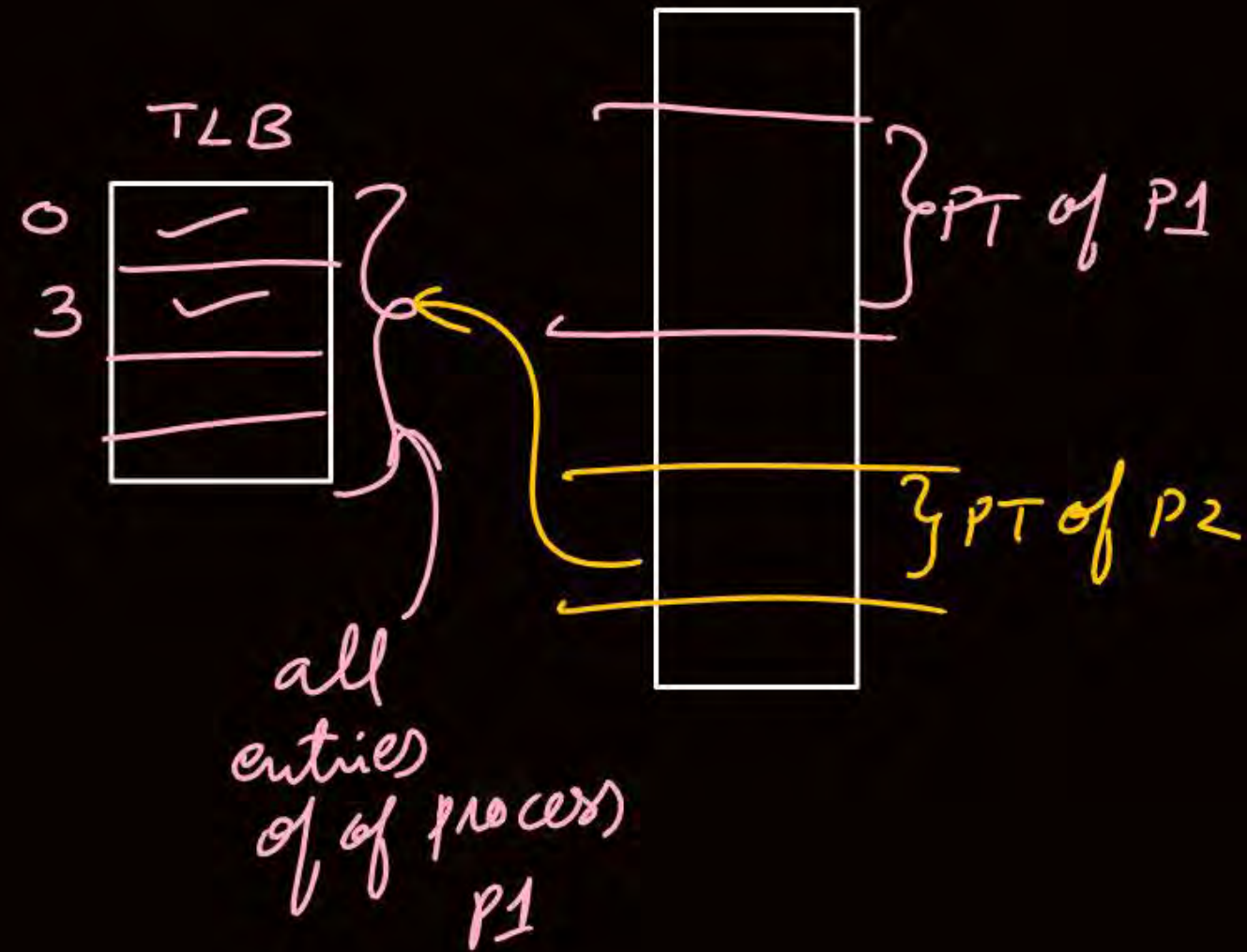
- Segmentation suffers from external fragmentation



what happens to TLB entries when process context switch happens?



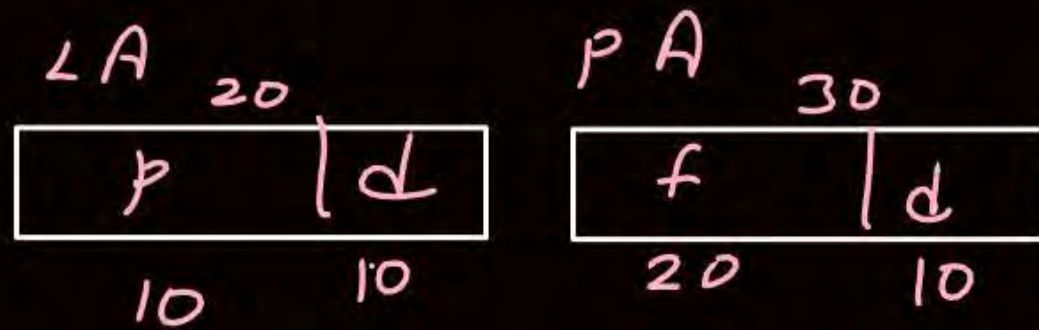
TLB is emptied



what happens when page size increases? $\Rightarrow$ page table size is decreased

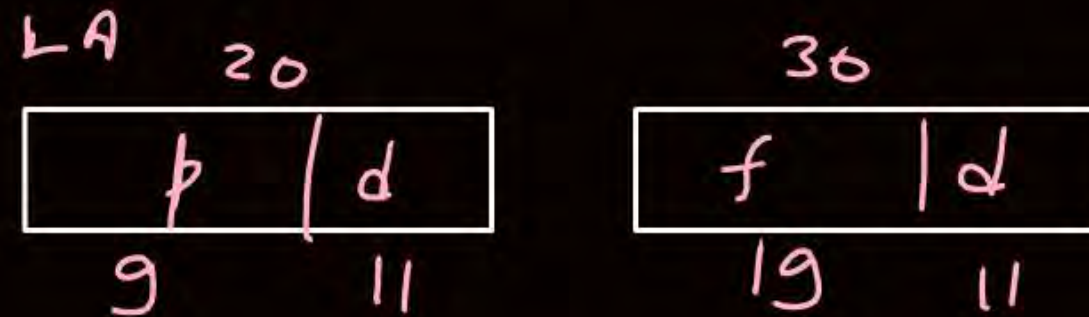
L.A. = 20 bits P.A. = 30 bits

if page size = 1KB



P.T. Size = $2^{10} * 20$ bits

Page size = 2KB



P.T. Size = $2^9 * 19$ bits



2 mins Summary

Topic

TLB

Topic

TLB Mapping

Topic

Segmentation



Happy Learning

THANK - YOU

